

ANATOMY OF THE NERVOUS SYSTEM

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DR OKESINA AKEEM AYODEJI

Nervous system

Central nervous system (CNS)

Brain

Spinal cord
Connects brain and peripheral nervous system

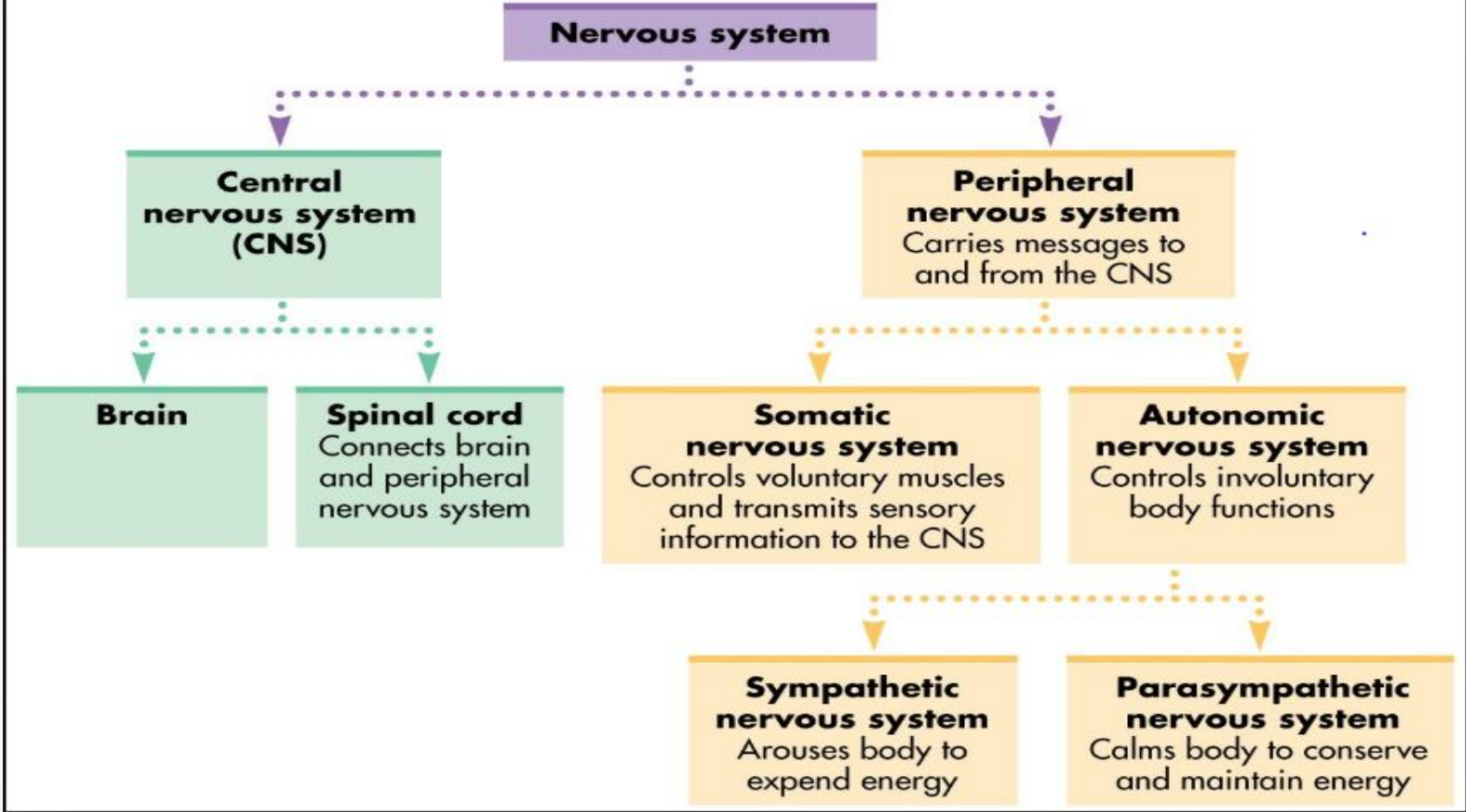
Peripheral nervous system
Carries messages to and from the CNS

Somatic nervous system
Controls voluntary muscles and transmits sensory information to the CNS

Autonomic nervous system
Controls involuntary body functions

Sympathetic nervous system
Arouses body to expend energy

Parasympathetic nervous system
Calms body to conserve and maintain energy



Nervous System is the most complex system in the human body, is formed by a network of more than 100 million nerve cells (**neurons**), assisted by many more glial cells.

Each neuron has, on average, at least 1000 interconnections with other neurons, forming a very complex system for communication.

Nerve tissue is distributed throughout the body as an integrated communications network.

Anatomically, the nervous system is divided into the central nervous system consisting of the brain and the spinal cord, and

the peripheral nervous system composed of nerve fibers and small aggregates of nerve cells called nerve ganglia

Neurons - responsible for conduction, propagation, and reception of nervous impulses.

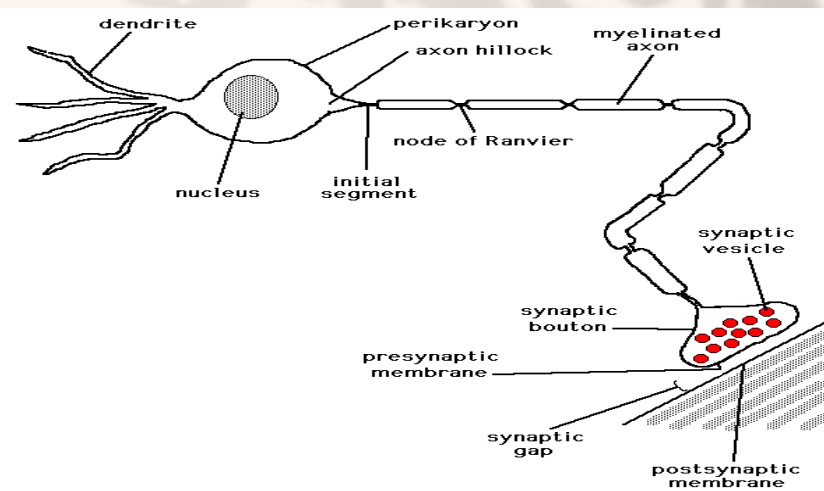
Processes called axons or dendrites extend from these cells.

Glial cells - (neuroglia) cells associated with neurons. No axons or dendrites. These cells are involved in nutrition, support, insulation, protection of neurons.

NEURONS

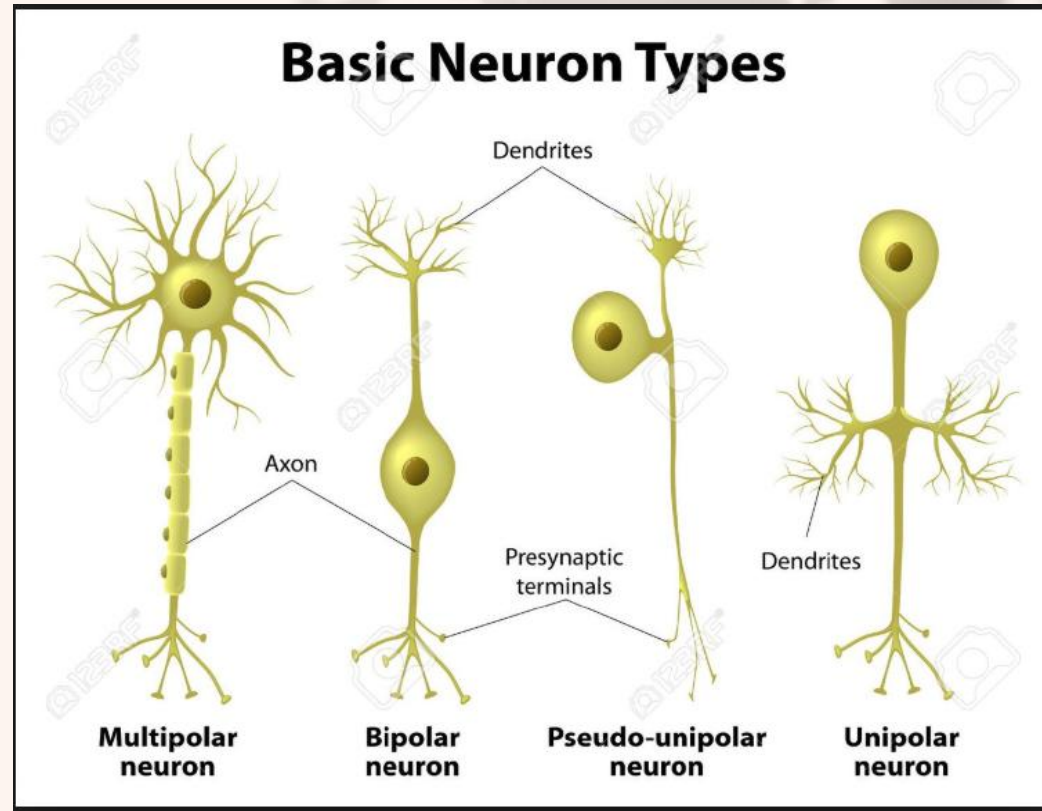
Most neurons consist of three parts:

- The **dendrites**, which are multiple elongated processes specialized in receiving stimuli from the environment, sensory epithelial cells, or other neurons.
- The **cell body**, or **perikaryon** which is the trophic center for the whole nerve cell and is also receptive to stimuli; and
- The **axon** which is a single process specialized in generating or conducting nerve impulses to other cells (nerve, muscle, and gland cells).
- Axons may also receive information from other neurons; this information mainly modifies the transmission of action potentials to other neurons
- The distal portion of the axon is usually branched and constitutes the **terminal arborization**. Each branch of this arborization terminates on the next cell in dilatations called **end bulbs (boutons)**, which interact with other neurons or non-nerve cells, forming structures called **synapses**. Synapses transmit information to the next cell in the circuit.



Based on number of processes found on cell body

1. **multipolar** = several dendrites & one axon
 - most common cell type
2. **bipolar** neurons = one main dendrite & one axon
 - found in retina, inner ear & olfactory
3. **unipolar** neurons = one process only(develops from a bipolar)
 - are always sensory neurons



Neurons can also be classified according to their functional roles.

Motor (efferent) neurons control effector organs such as muscle fibers, exocrine and endocrine glands.

Sensory (afferent) neurons are involved in the reception of sensory stimuli from the environment and from within the body.

Interneurons establish relationships among other neurons, forming complex functional networks or circuits (as in the retina).

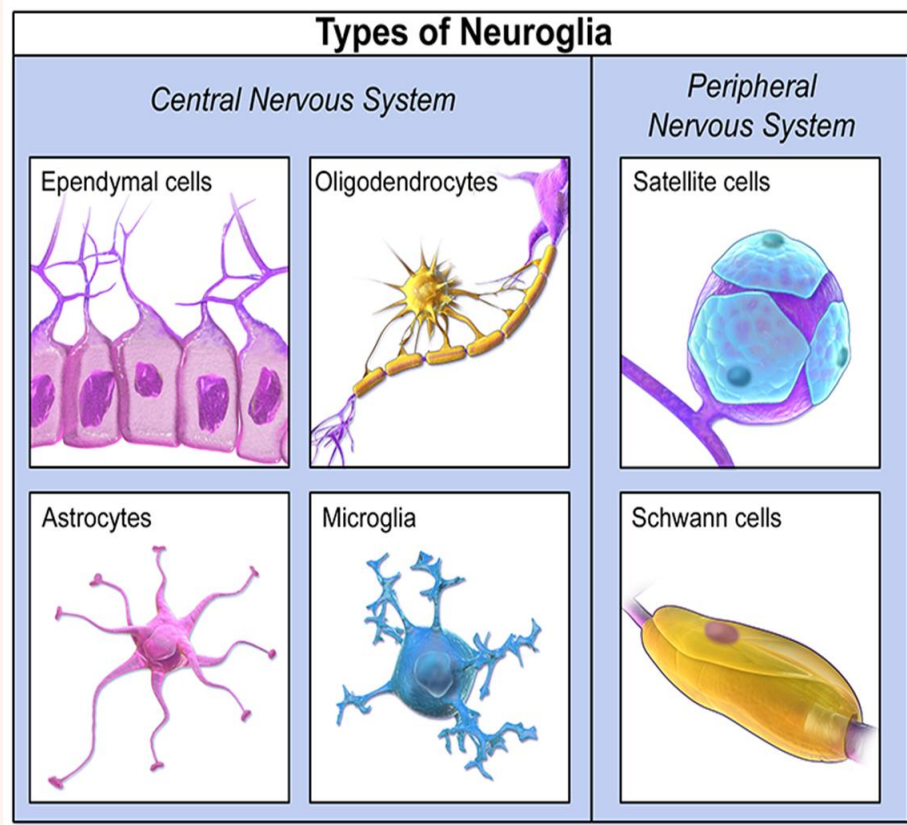
GLIAL CELLS

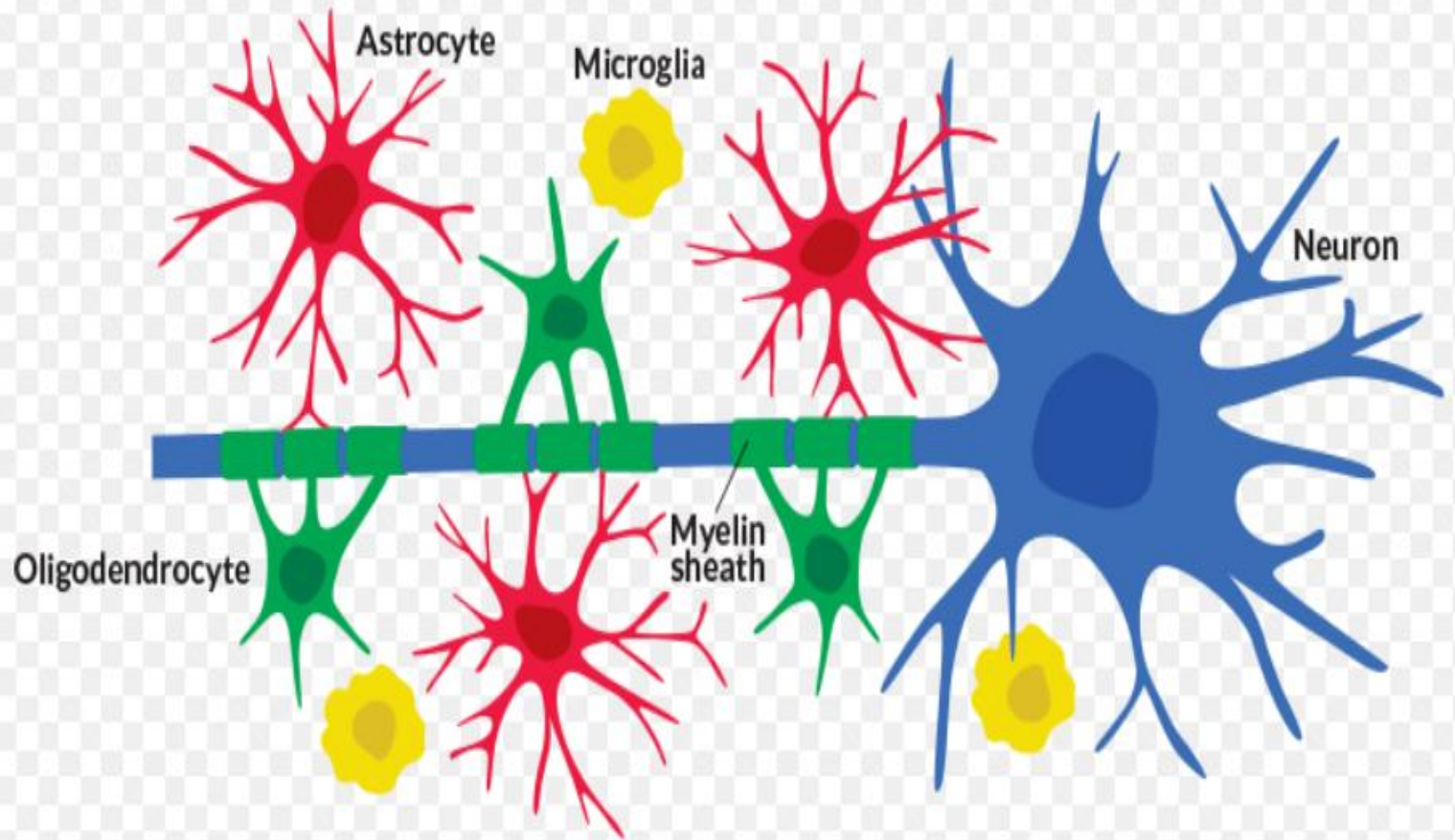
There Are Six Major Types
Of Glial Cells Which Are
CNS

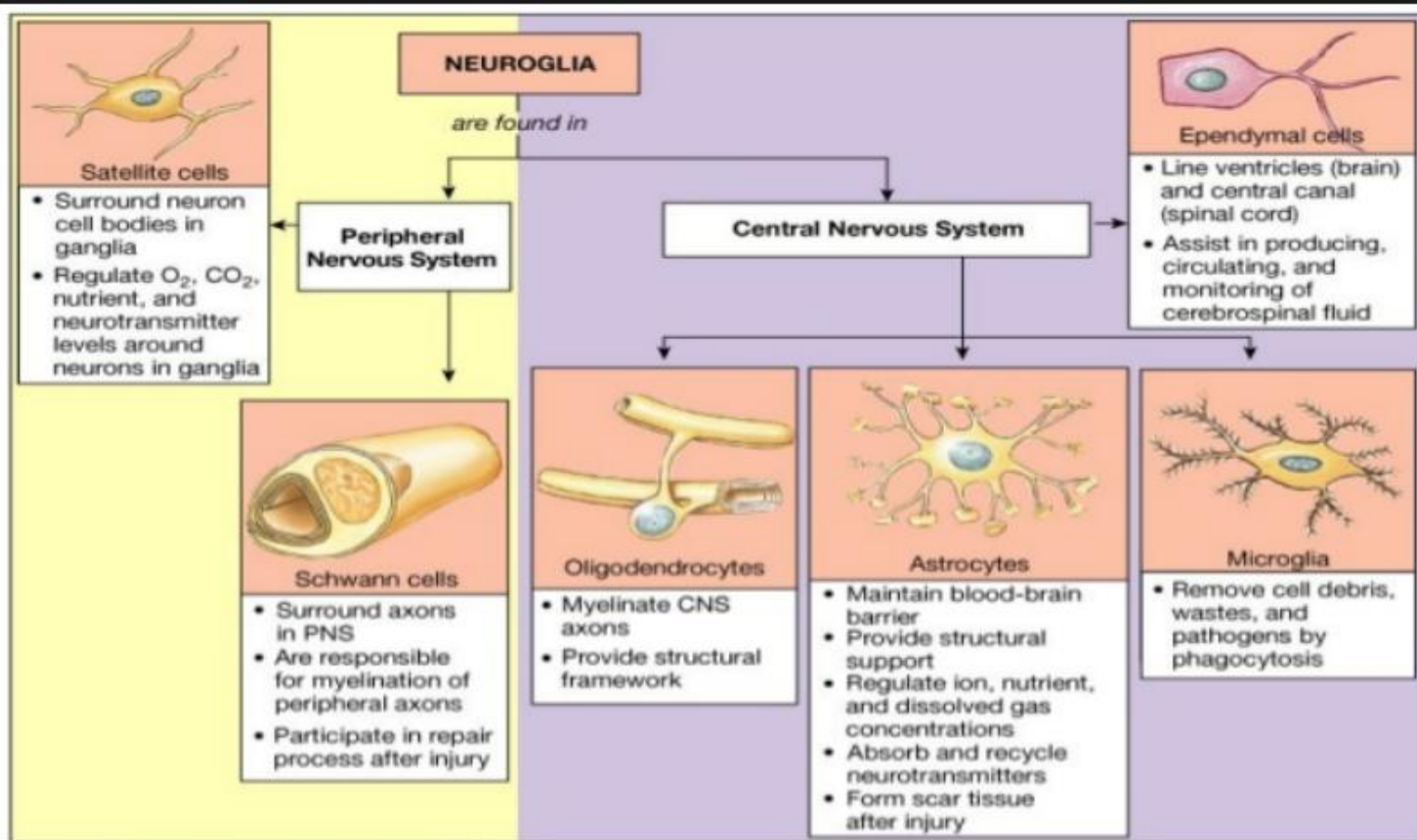
1. Astrocyte
2. Oligodendrocyte
3. Microglia
4. Ependymal Cells

PNS

1. Schwann Cells
2. Satellite Cells







Central Nervous System (CNS)

The CNS consists of the **brain** and **spinal cord**. It is responsible for integration, processing, and control.

A. Brain

The brain is located within the cranial cavity and weighs approximately 1.3–1.4 kg in adults.

Major Parts of the Brain

i. **Cerebrum** ; Largest part of the brain, Divided into right and left hemispheres, Each hemisphere has lobes:

- **Frontal lobe**: voluntary movement, speech, personality
- **Parietal lobe**: sensory perception
- **Temporal lobe**: hearing, memory
- **Occipital lobe**: vision

Outer layer: **cerebral cortex** (gray matter)

Inner region: **white matter**

ii. Diencephalon

Located between cerebrum and brainstem

Components:

- **Thalamus**: relay station for sensory impulses
- **Hypothalamus**: temperature regulation, hunger, autonomic control, endocrine functions
- **Epithalamus**: includes pineal gland

iii. Brainstem

Connects the brain to the spinal cord.

Midbrain: visual and auditory reflexes

Pons: bridge between cerebrum and cerebellum

Medulla oblongata: vital centers for respiration, heart rate, and blood pressure

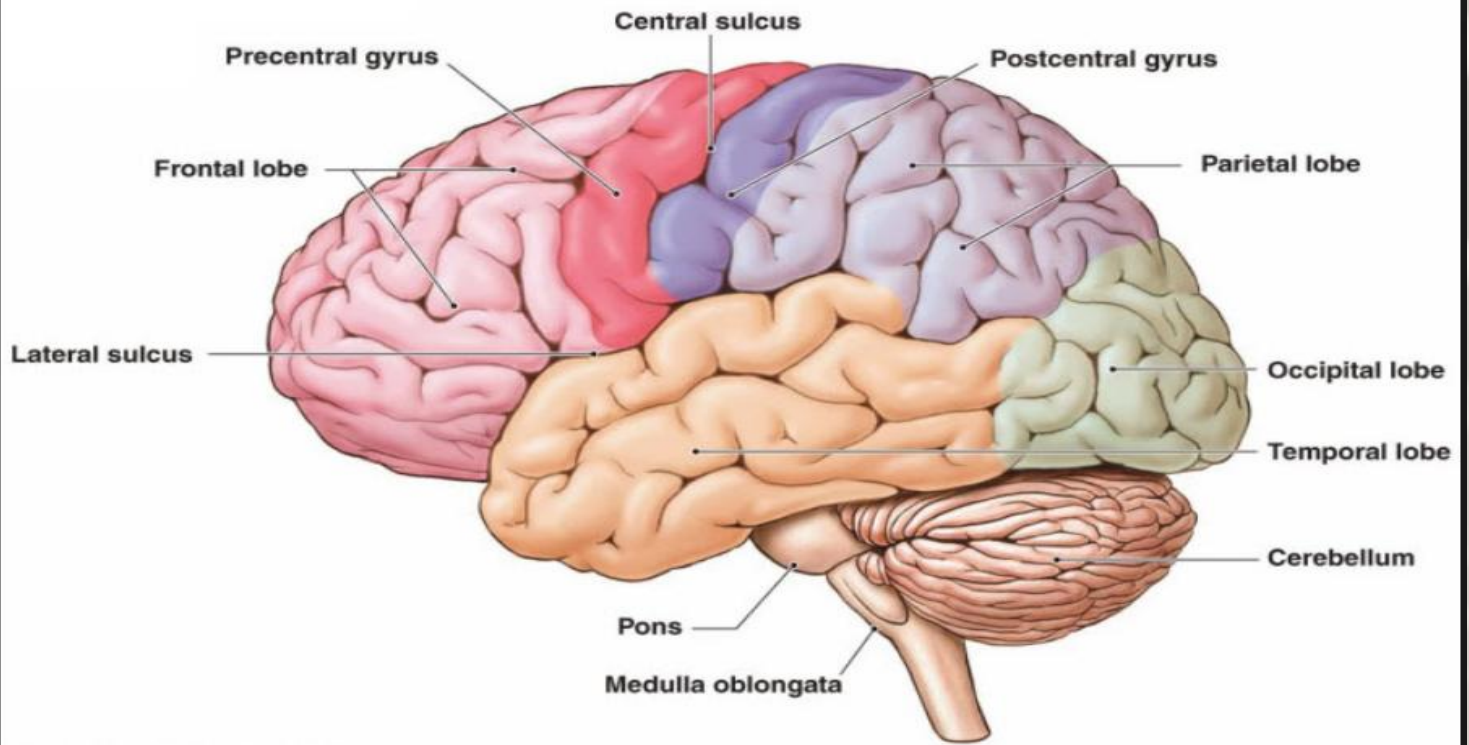
iv. Cerebellum

Located posterior to the brainstem

Coordinates voluntary movement

Maintains posture, balance, and muscle tone

Brain



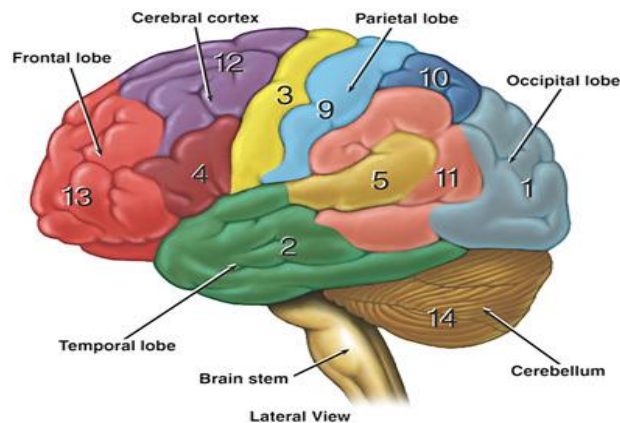
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Functional Areas of the Cerebral Cortex

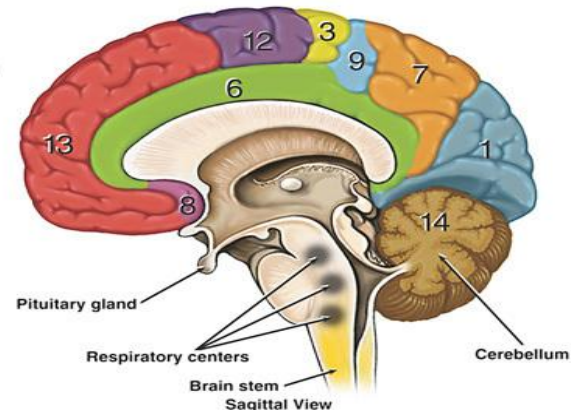
- 1 **Visual Area:**
Sight
Image recognition
Image perception
- 2 **Association Area**
Short-term memory
Equilibrium
Emotion
- 3 **Motor Function Area**
Initiation of voluntary muscles
- 4 **Broca's Area**
Muscles of speech
- 5 **Auditory Area**
Hearing
- 6 **Emotional Area**
Pain
Hunger
"Fight or flight" response
- 7 **Sensory Association Area**
- 8 **Olfactory Area**
Smelling
- 9 **Sensory Area**
Sensation from muscles and skin
- 10 **Somatosensory Association Area**
Evaluation of weight, texture, temperature, etc. for object recognition
- 11 **Wernicke's Area**
Written and spoken language comprehension
- 12 **Motor Function Area**
Eye movement and orientation
- 13 **Higher Mental Functions**
Concentration
Planning
Judgment
Emotional expression
Creativity
Inhibition

Functional Areas of the Cerebellum

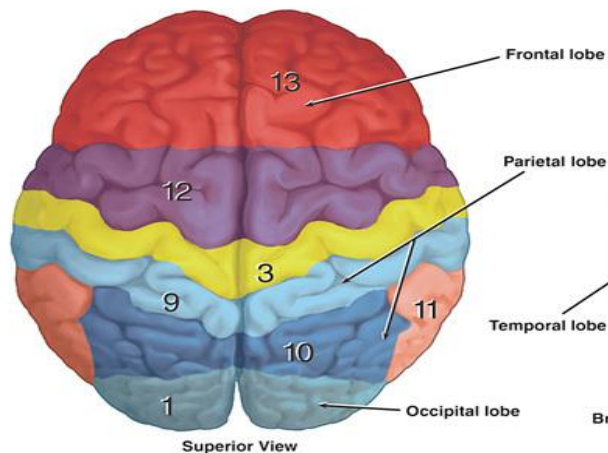
- 14 **Motor Functions**
Coordination of movement
Balance and equilibrium
Posture



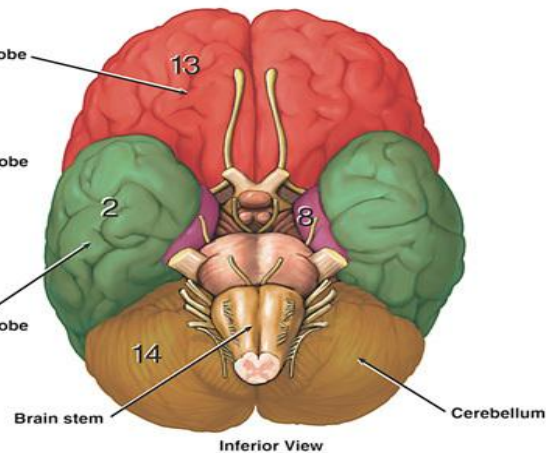
Lateral View



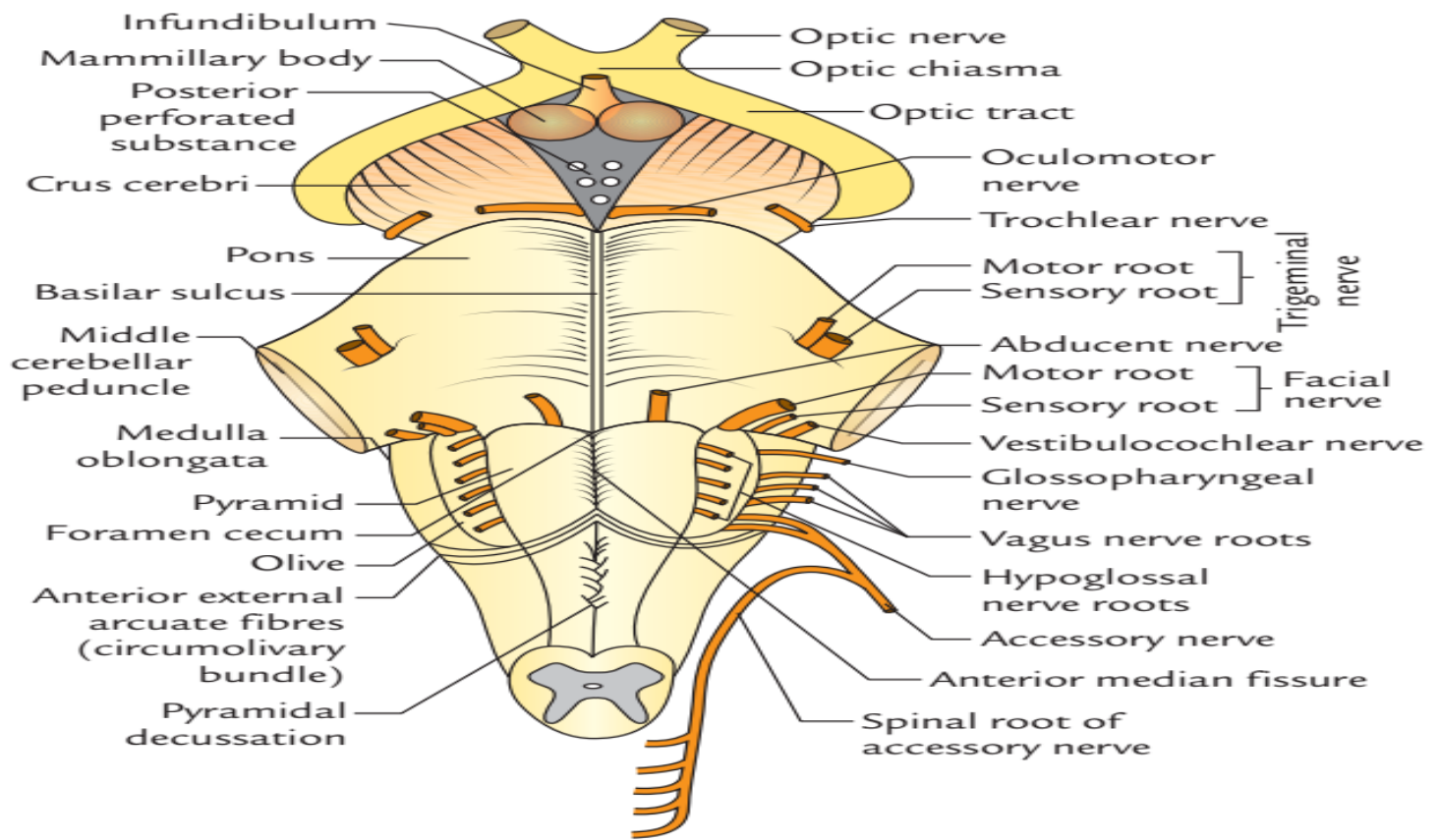
Sagittal View

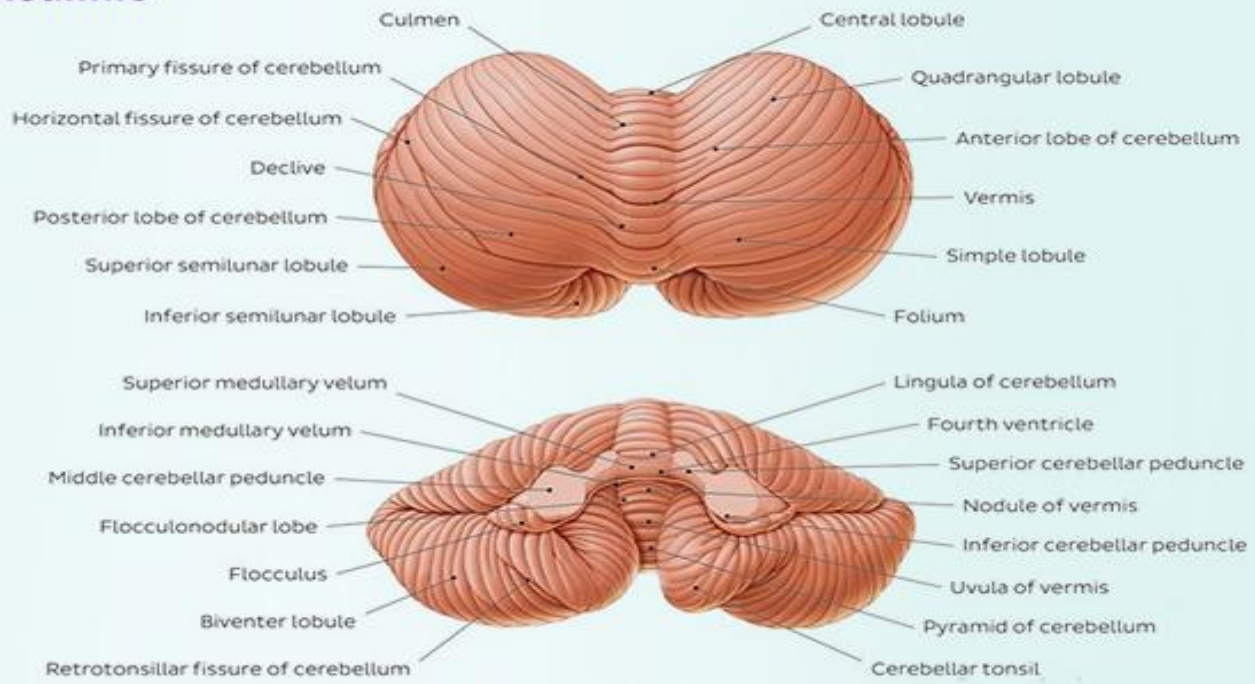


Superior View



Inferior View



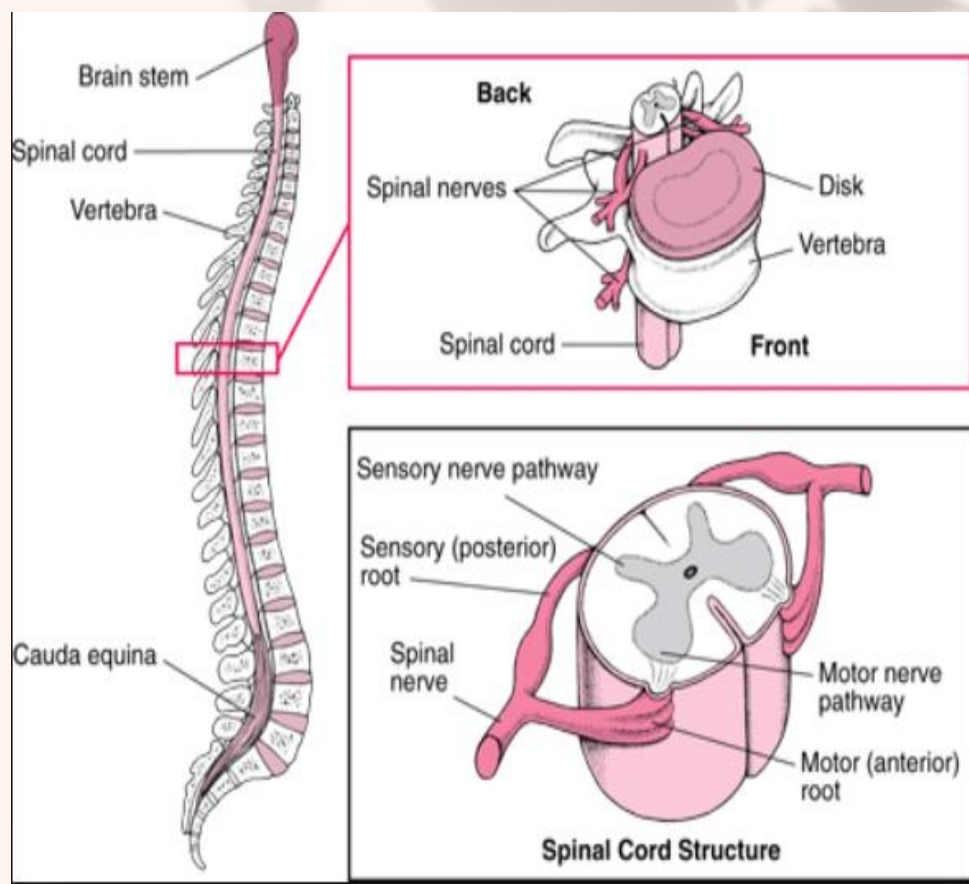


B. Spinal Cord:

- Extends from the foramen magnum to L1–L2 vertebral level
- Protected by vertebral column and meninges
- Shows cervical and lumbar enlargements
- Gives rise to 31 pairs of spinal nerves

Structure:

- Central gray matter (H-shaped) Surrounding white matter
- Central canal contains cerebrospinal fluid (CSF)



MENINGES

The **central nervous system (CNS)** is protected by the **skull and vertebral column** and further enclosed by three connective tissue membranes collectively called the **meninges**. From superficial to deep, these are the **dura mater**, **arachnoid mater**, and **pia mater**. The arachnoid and pia mater are closely related and together form the **pia-arachnoid**.

1. Dura Mater: Outermost, thick, dense connective tissue layer

Continuous with the **periosteum of the skull**

In the spinal cord, separated from vertebrae by the **epidural space**, which contains:

- Thin-walled veins, Loose connective tissue and Adipose tissue

Always separated from the arachnoid by the **subdural space**

Surfaces lined by **simple squamous epithelium** of mesenchymal origin

2. Arachnoid Mater

Middle, delicate membrane

Composed of:

- An outer layer adjacent to the dura mater
- **Trabeculae** connecting it to the pia mater

Spaces between trabeculae form the **subarachnoid space**

- Filled with **cerebrospinal fluid (CSF)**
- Communicates with brain ventricles
- Acts as a **hydraulic cushion** protecting the CNS

Avascular (contains no blood vessels)

Lined by simple squamous epithelium

Forms **arachnoid villi**, which project into venous sinuses

- Function: **reabsorption of CSF into venous blood**

3. Pia Mater

- Innermost, thin, highly vascular membrane
- Composed of loose connective tissue
- Closely follows the contours of the brain and spinal cord
- Does not directly contact neural tissue
 - Separated by a thin layer of neuroglial processes forming a physical barrier
- Blood vessels enter the CNS through perivascular spaces covered by pia mater
- Pia mater disappears before vessels become capillaries
- Covered by squamous cells of mesenchymal origin

4. Choroid Plexus

Formed by invaginations of pia mater into brain ventricles

Rich in dilated, fenestrated capillaries

Located in:

- Roofs of the **third and fourth ventricles**
- Part of the **lateral ventricles**

Covered by **simple cuboidal or low columnar epithelium**

Primary function: **production of cerebrospinal fluid (CSF)**

5. Cerebrospinal Fluid (CSF)

Clear, colorless fluid with:

- Low density (1.004–1.008 g/mL)
- Very low protein content
- Few cells (2–5 lymphocytes/mL)

Circulates through:

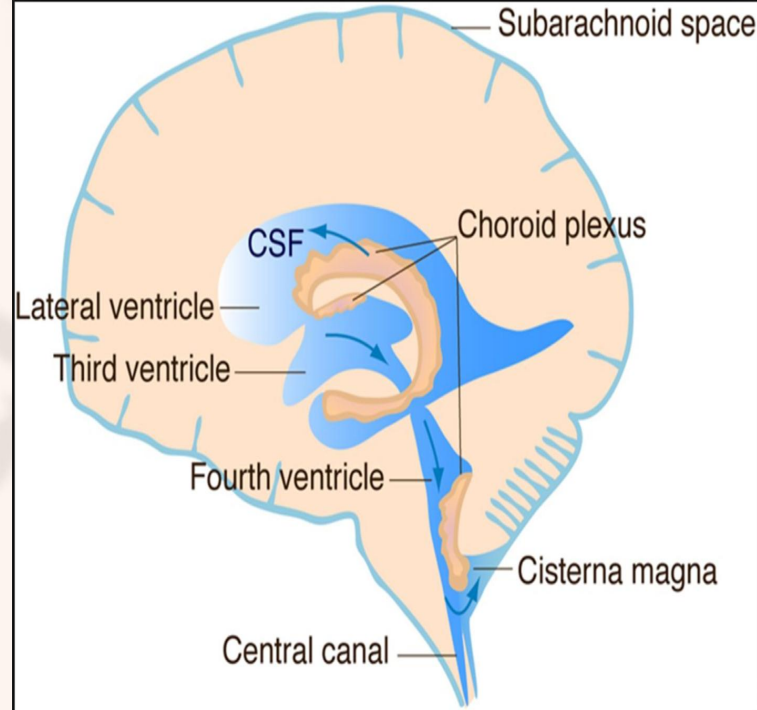
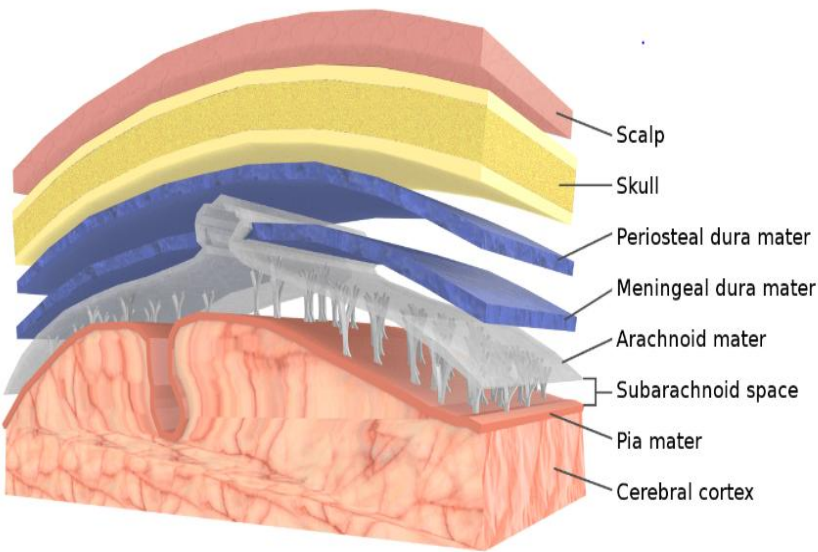
- Brain ventricles
- Central canal of spinal cord
- Subarachnoid and perivascular spaces

Functions:

- Nutritional and metabolic support of CNS
- Mechanical protection against shock

Continuously produced by choroid plexus

Absorbed mainly through **arachnoid villi** into venous circulation



PERIPHERAL NERVOUS SYSTEM

The main components of the peripheral nervous system are the **nerves**, **ganglia**, and **nerve endings**. Nerves are bundles of nerve fibers surrounded by connective tissue sheaths.

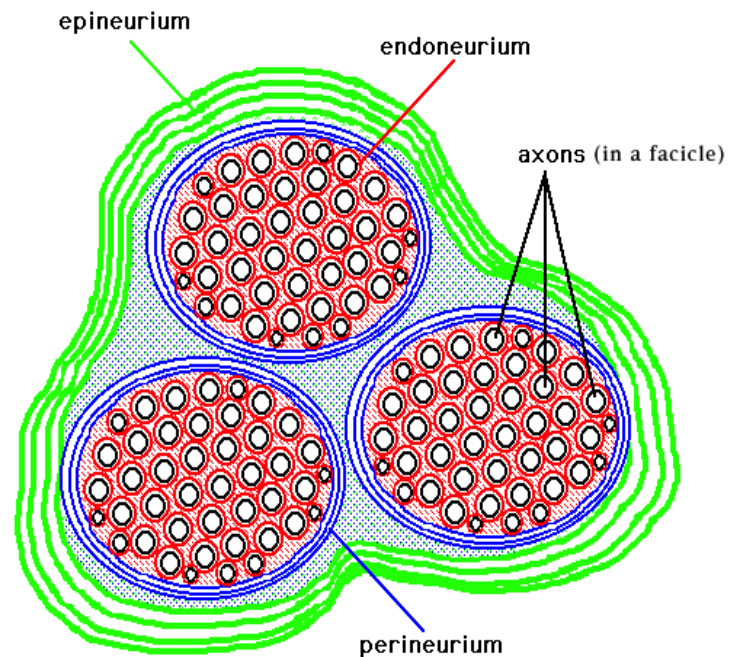
STRUCTURE OF THE NERVE

The nerves have an external fibrous coat of dense connective tissue called **epineurium**, which also fills the space between the bundles of nerve fibers.

Each bundle is surrounded by the **perineurium**, a sleeve formed by layers of flattened epitheliumlike cells.

Within the perineurial sheath run the Schwann cell-sheathed axons and their enveloping connective tissue, the **endoneurium**.

The endoneurium consists of a thin layer of reticular fibers, produced by Schwann cells.



GANGLIA

Ganglia are ovoid collections of **neuronal cell bodies** located outside the central nervous system. They also contain **glial (satellite) cells** and are generally supported by **connective tissue**. Functionally, ganglia act as **relay stations** in peripheral nerve pathways, with **one nerve entering and another exiting** each ganglion.

Anatomically, ganglia are classified based on the **direction of nerve impulse transmission** into **sensory ganglia** and **autonomic ganglia**.

1. Sensory Ganglia

Contain neuronal cell bodies of **afferent (sensory) neurons**

Transmit impulses **toward the CNS**

Two anatomical types:

- **Cranial sensory ganglia:** associated with cranial nerves
- **Spinal (dorsal root) ganglia:** associated with the dorsal roots of spinal nerves

Typically enclosed by a **connective tissue capsule**

Neurons are usually **pseudounipolar**

Cell bodies are surrounded by **satellite glial cells**

Autonomic Ganglia

Part of the **autonomic nervous system**

Appear as **bulbous enlargements** along autonomic nerves

Function as relay stations between **preganglionic** and **postganglionic neurons**

Some autonomic ganglia are located within organs, especially:

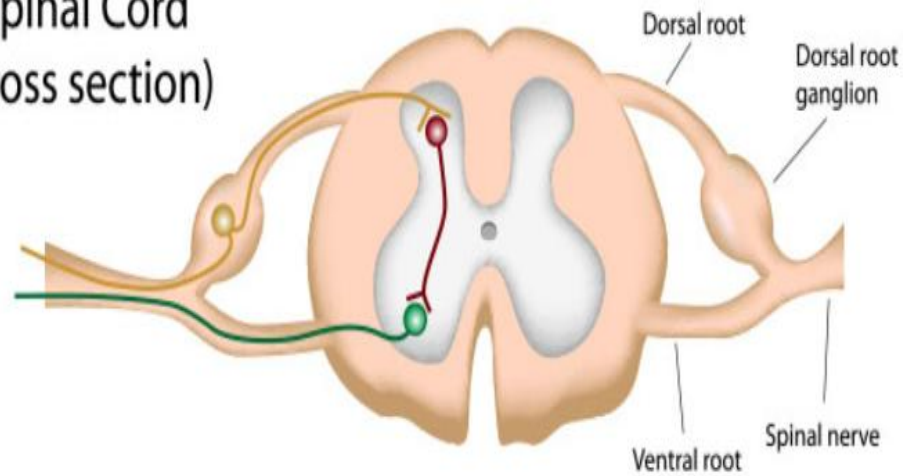
- Walls of the **gastrointestinal tract**
- Known as **intramural ganglia**

Intramural ganglia:

- Lack a connective tissue capsule
- Neuronal cells are supported by the **stromal tissue of the organ**

Neurons are typically **multipolar**

Spinal Cord (cross section)



The somatic nervous system (SNS or voluntary nervous system)

It is the part of the peripheral nervous system associated with the **voluntary control of body movements via skeletal muscles**.

The somatic nervous system consists of **afferent nerves or sensory nerves, and efferent nerves or motor nerves**.

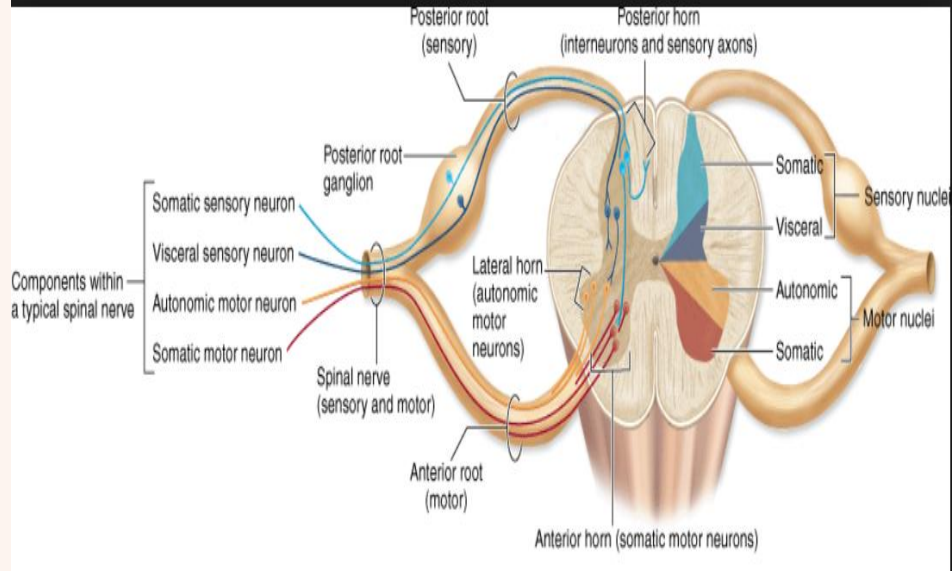
Afferent nerves are responsible for relaying sensation from the body to the central nervous system; **efferent nerves** are responsible for sending out commands from the CNS to the body, stimulating muscle contraction; they include all the non-sensory neurons connected with skeletal muscles and skin.

There are **43 segments** of nerves in the human body. With each segment, there is a pair of **sensory and motor nerves**. In the body, **31 segments of nerves are in the spinal cord** and **12 are in the brain stem**.

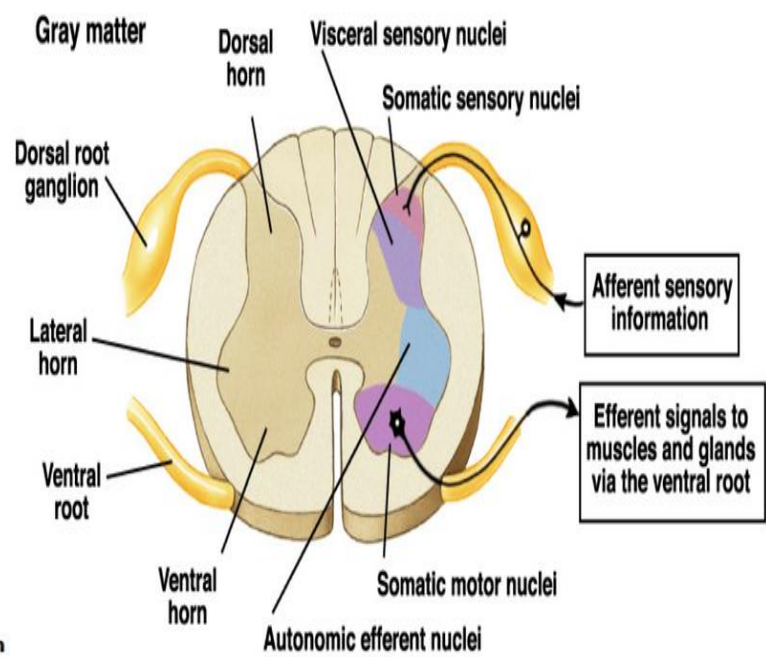
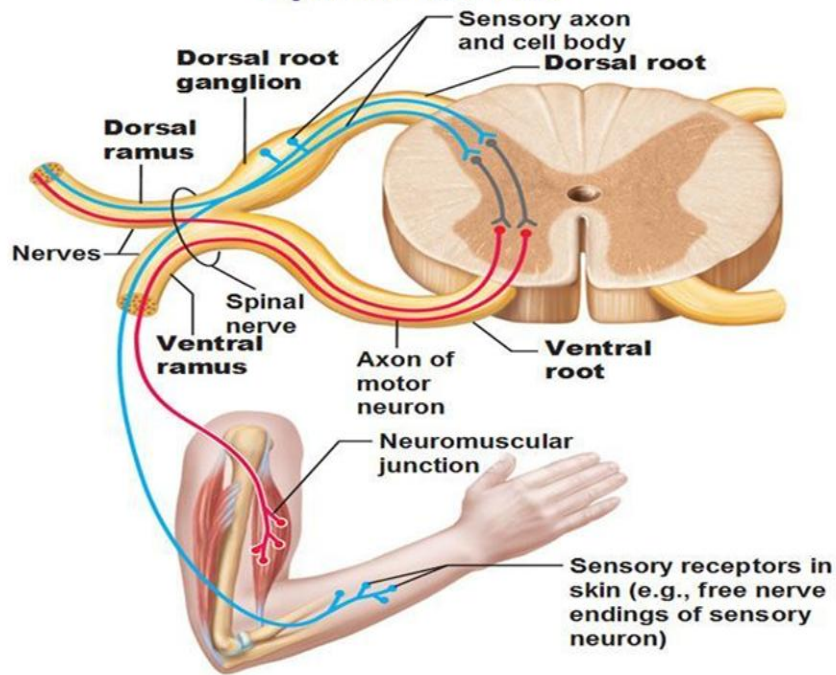
Thus, the somatic nervous system consists of two parts:

Spinal nerves: They are peripheral nerves that carry sensory information into and motor commands out of the spinal cord.

Cranial nerves: They are the nerve fibers that carry information into and out of the brain stem. They include smell, vision, eye, eye muscles, mouth, taste, ear, neck, shoulders, and tongue



Spinal Nerves



Classification of cranial nerves

Sensory cranial nerves: contain only afferent (sensory) fibers

- **I** Olfactory nerve
- **II** Optic nerve
- **VIII** Vestibulocochlear nerve

Motor cranial nerves: contain only efferent (motor) fibers

- **III** Oculomotor nerve
- **IV** Trochlear nerve
- **VI** Abducent nerve
- **XI** Accessory nerve
- **XII** Hypoglossal nerve

Mixed nerves: contain both sensory and motor fibers---

- **V** Trigeminal nerve,
- **VII** Facial nerve,
- **IX** Glossopharyngeal nerve
- **X** Vagus nerve

DEFINATION TERMS IN AUTONOMIC NERVOUS SYSTEM

Ganglion: a structure containing several nerve cell bodies, typically linked by synapses, and often forming a swelling on a nerve fibre

Synapses: also called **neuronal junction**, the site of transmission of electric nerve impulses between two nerve cells (neurons) or between a neuron and a gland or muscle cell (effector).

A synaptic connection between a neuron and a muscle cell is called a ***neuromuscular junction***.

Preganglionic neurons: **proximal to a ganglion specifically:** of, relating to, or being a usually myelinated axon arising from a cell body in the central nervous system and terminating in an autonomic ganglion

Postganglionic neurons: A neuron of the autonomic nervous system whose cell body lies in an autonomic ganglion and whose axon terminates in a visceral effector (smooth or cardiac muscle or glands).

Presynaptic neurons: is a neuron (nerve cell) that fires the neurotransmitter because of an action potential entering its axon terminal. In both the central and peripheral nervous systems in mammals, presynaptic terminals operate mostly in the same way

Postsynaptic neuron: is a neuron (nerve cell) that receives the neurotransmitter after it has crossed the synapse and may experience an action potential if the neurotransmitter is strong enough. Postsynaptic neurons work through temporal summation and spatial summation

Neurotransmitter: is a signalling molecule secreted by a neuron or a glial cell to affect another cell across a synapse. The cell receiving the signal, or target cell, may be another neuron, but could also be a gland or muscle cell

Autonomic Nervous System

The **autonomic nervous system (ANS)** is a division of the **peripheral nervous system** responsible for the **involuntary control of smooth muscle, cardiac muscle, and glands**, thereby regulating the functions of internal organs (viscera). It operates largely **without conscious control** and maintains internal homeostasis.

Anatomical Organization of the ANS

Central Component

Consists of autonomic control centers located in:

- **Hypothalamus** (principal regulatory center)
- **Brainstem**
- **Spinal cord**

Integrates visceral sensory information and coordinates autonomic output

Peripheral Component

Composed of:

- **Autonomic nerves**
- **Autonomic ganglia**
- **Pre- and postganglionic fibers**

Forms part of the **peripheral nervous system**

Functional Components of the ANS

Visceral Afferent Fibers

Carry sensory impulses from internal organs to the CNS

Provide feedback for autonomic reflexes

Visceral Efferent Fibers

Transmit motor impulses from CNS to visceral organs

Organized as a **two-neuron chain**:

- **Preganglionic neuron**: cell body in CNS
- **Postganglionic neuron**: cell body in autonomic ganglion

Divisions of the ANS

1. Sympathetic Nervous System (SNS)

Known as the **fight-or-flight** system

Thoracolumbar division

- Originates from T1–L2 spinal cord segments

Preganglionic neurons synapse in:

- Sympathetic chain ganglia
- Prevertebral ganglia

Generally, prepares the body for action

Neurotransmitters:

- Preganglionic: Acetylcholine (ACh)
- Postganglionic: Norepinephrine (NE)

2. Parasympathetic Nervous System (PNS)

Known as the **rest-and-digest** system

Craniosacral division

- Originates from brainstem nuclei and S2–S4 spinal segments

Ganglia located close to or within target organs

Promotes energy conservation and maintenance

Neurotransmitters:

- Preganglionic: Acetylcholine (ACh)
- Postganglionic: Acetylcholine (ACh)

3. Enteric Nervous System (ENS)

Extensive intrinsic network within the **gastrointestinal tract**

Contains over **100 million neurons**

Organized into:

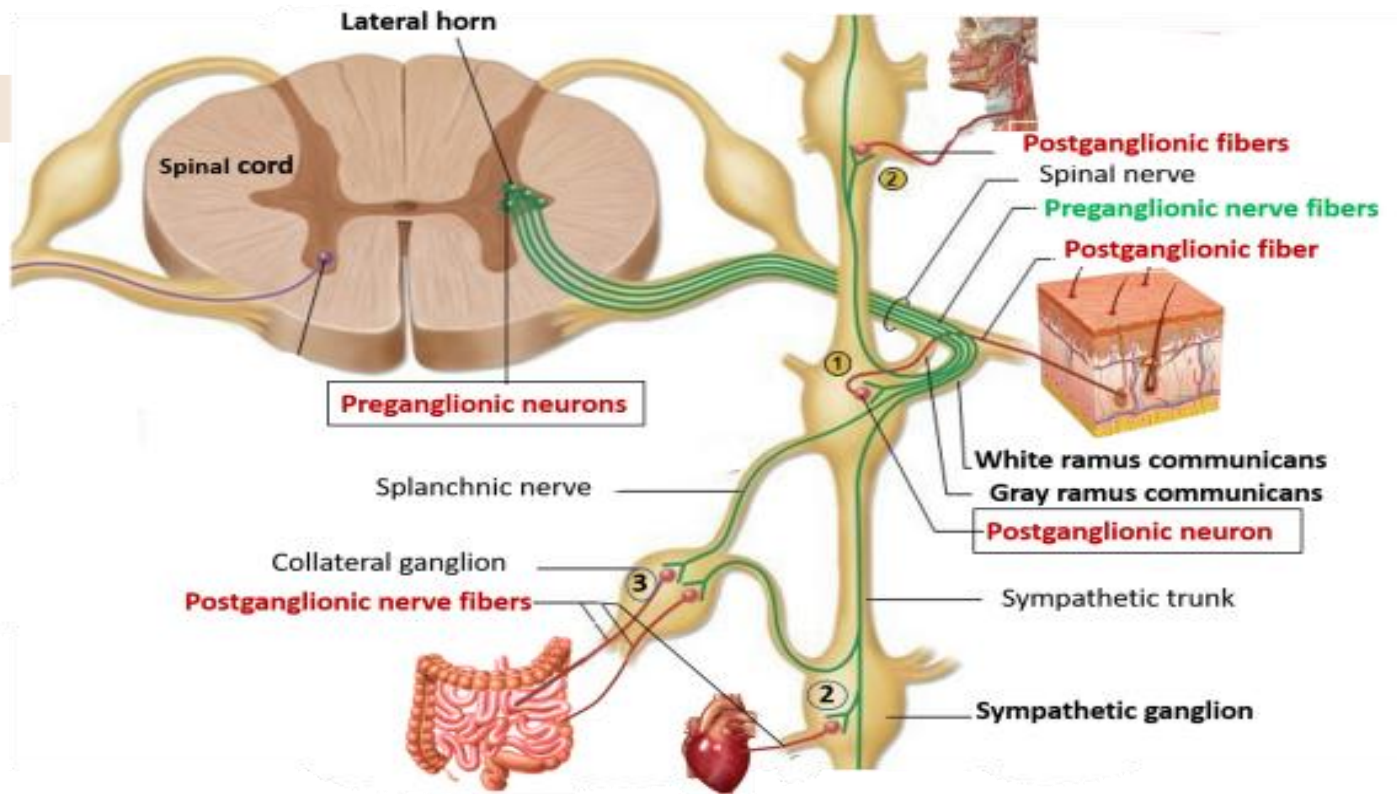
- Myenteric (Auerbach) plexus
- Submucosal (Meissner) plexus

Capable of **independent function**

Regulates digestion, secretion, and gut motility

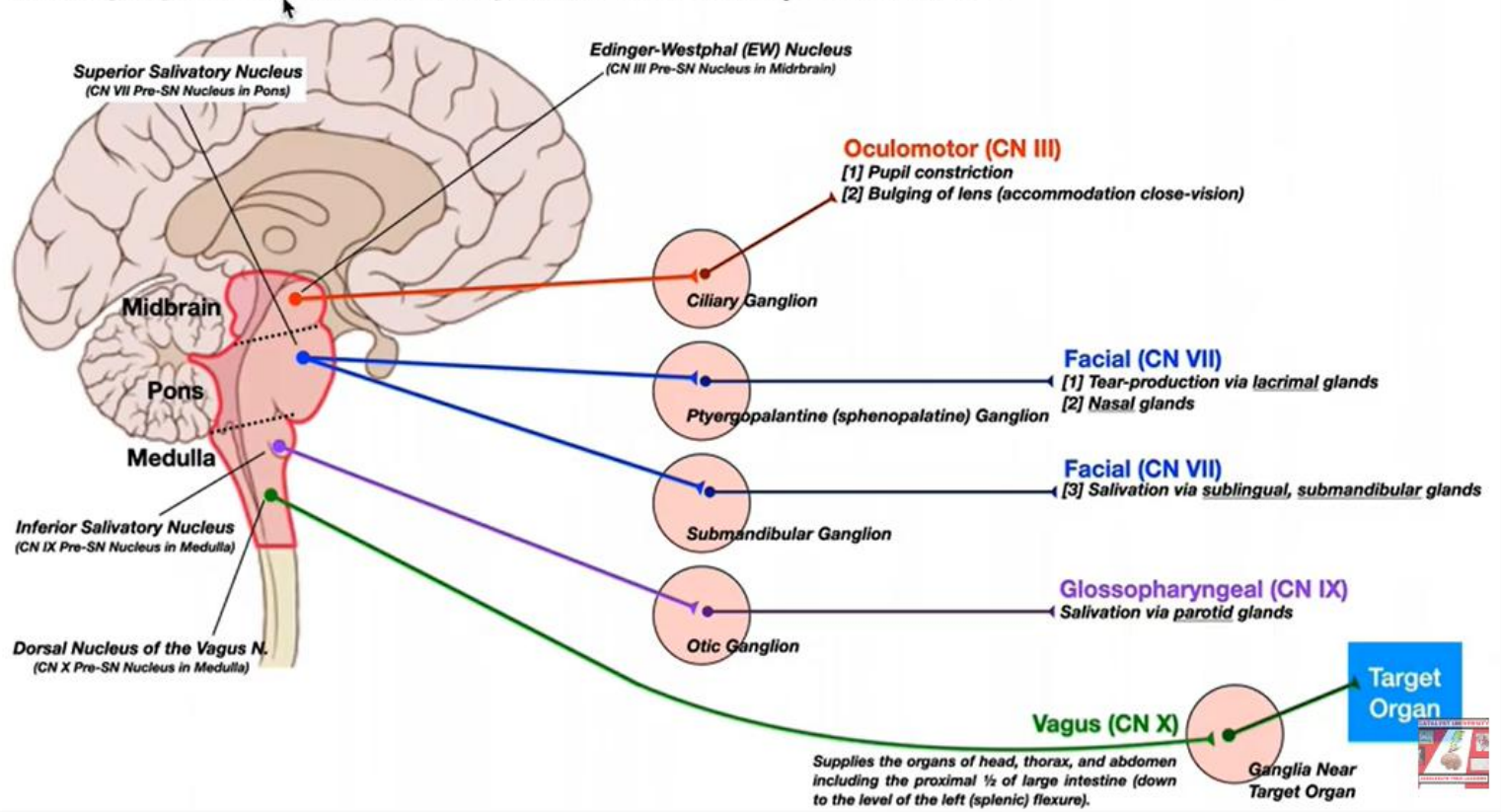
Neurotransmitters include:

- Acetylcholine
- Nitric oxide
- Serotonin

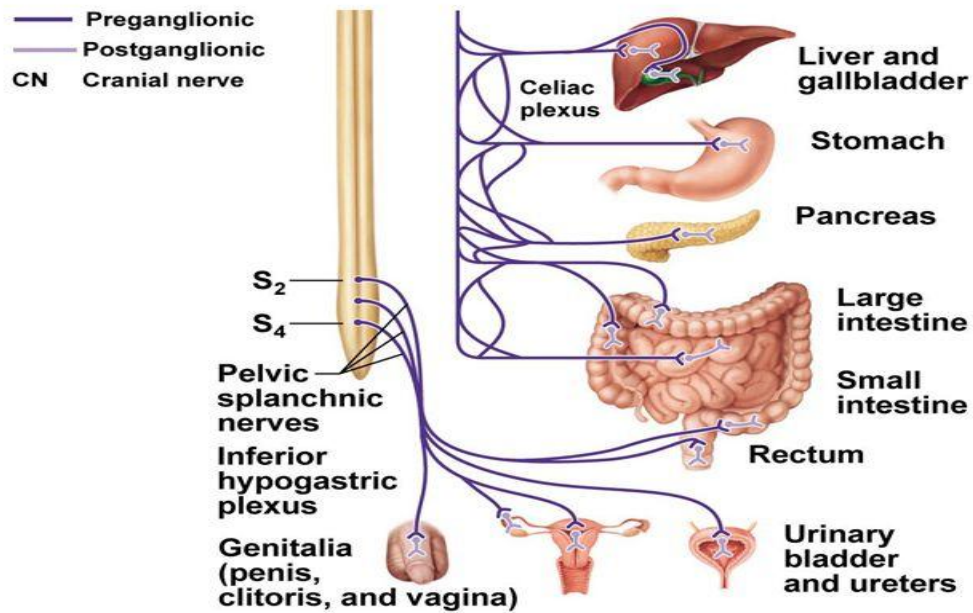


Sympathetic Nervous System (SNS)

Parasympathetic Nervous System Overview | Cranial Part



Parasympathetic Nervous System (sacral outflow)



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Characteristics of ANS

Sympathetic Effects

Fight or flight response. Release of norepinephrine (NT) from postganglionic fibers and epinephrine (NT) from adrenal medulla. Mass activation prepares for intense activity. Heart rate (HR) increases. Bronchioles dilate. Blood [glucose] increases.

Parasympathetic Effects

Normally not activated as a whole. Stimulation of separate parasympathetic nerves. Release ACh as NT. Relaxing effects: Decreases HR. Dilates visceral blood vessels. Increases digestive activity.

Autonomic Functions

The ANS regulates:

Heart rate and cardiac output

Vascular tone and blood pressure

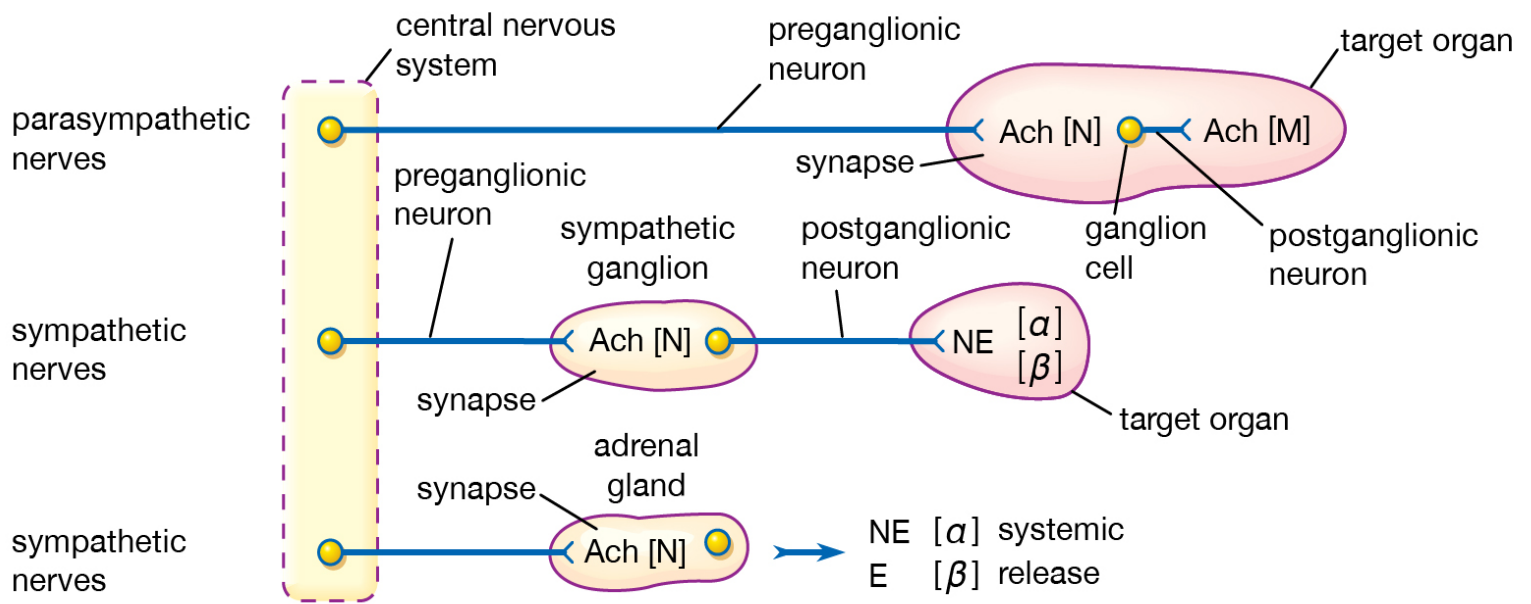
Respiratory rate

Gastrointestinal motility and secretion

Pupillary diameter

Urination and defecation

Glandular secretion



Ach	Acetylcholine	[N]	Nicotinic acetylcholine receptor
E	Epinephrine	[M]	Muscarinic acetylcholine receptor
NE	Norepinephrine	α	Norepinephrine receptor subtype
		β	Norepinephrine receptor subtype

Effects of the two branches of the ANS

Organ	Sympathetic Effect	Parasympathetic Effect
Pupil	dilation	constriction
Lens	Far focus (lower curvature)	Near focus (increased curvature)
Salivary Gland secretion	High in viscosity	serous
Heart	Increased rate and pressure	Lower rate and pressure
Lungs	Dilation of respiratory passages	Constriction of respiratory passages
Gastrointestinal	Decreased motility	Increased motility
Kidneys	Decreased filtration rate	Increased filtration rate
Male genitalia	Ejaculation	Erection
Vascular smooth muscle	Variable depending on the neurotransmitter	Relaxation
Sweat glands	Increased activity	No innervation
Arteries to skeletal muscle	dilation	No innervation
Veins	Variable depending on the neurotransmitter	No innervation

PARASYMPATHETIC

NERVOUS SYSTEM

"rest and digest" response

- Constricts pupils
- Increases saliva production
- Decreases heart rate
- Indirectly dilates blood vessels
- Constricts bronchi
- Decreases blood flow to skeletal muscles
- Increases stomach and intestine motility, increases blood flow to GI tract, and increases digestive enzyme secretion
- No involvement
- No involvement
- Contracts bladder



SYMPATHETIC

NERVOUS SYSTEM

"fight or flight" response

- Dilates pupils to enhance far vision
- Decreases saliva production
- Increases heart rate
- Constricts blood vessels, increases blood pressure
- Dilates bronchi to allow more air into lungs
- Increases blood flow to skeletal muscles
- Reduces stomach and intestine motility, diverts blood flow away from GI tract, and reduces digestive enzyme secretion
- Release glucose to increase blood sugar
- Stimulate adrenal glands to secrete stress hormones
- Relaxes bladder

SUMMARY OF BLOOD SUPPLY TO THE BRAIN

The brain receives blood from two sources: **the internal carotid arteries**, which arise at the point in the neck where the common carotid arteries bifurcate, and **the vertebral arteries**.

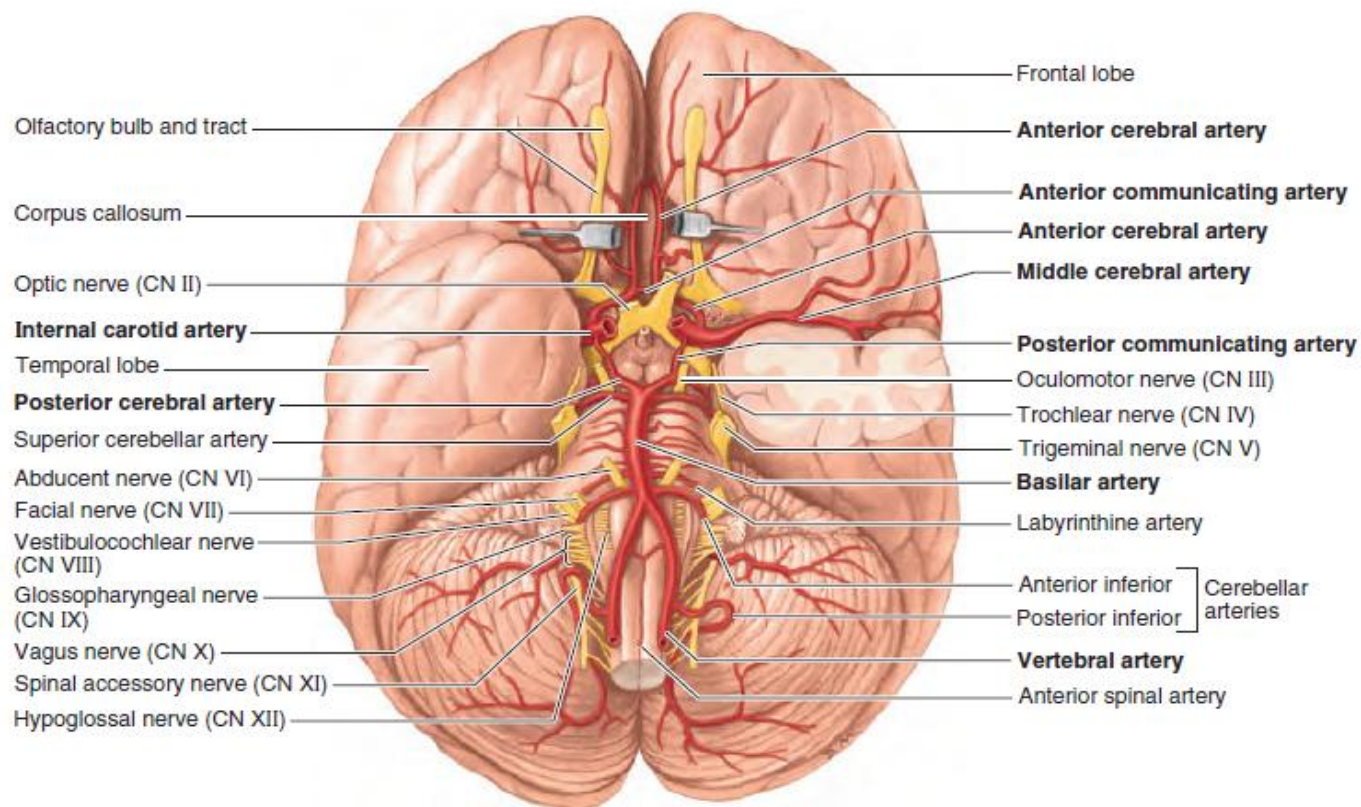
The internal carotid arteries branch to form two major cerebral arteries, the **anterior and middle cerebral arteries**.

The right and left vertebral arteries come together at the level of the **pons** on the ventral surface of the brainstem to form the **midline basilar artery**.

The basilar artery joins the blood supply from the internal carotids in an arterial ring at the base of the brain (in the vicinity of the hypothalamus and cerebral peduncles) **called the circle of Willis**.

The posterior cerebral arteries arise at this confluence, as do two small bridging arteries, **the anterior and posterior communicating arteries**.

Conjoining the two major sources of cerebral vascular supply via the circle of Willis presumably improves the chances of any region of the brain continuing to receive blood if one of the major arteries becomes occluded



Inferior view

Cerebral Arterial Circle

An important anastomoses at the base of the brain.

Formed by the several arteries that supply the brain.

Located in the inter-peduncular fossa in the base of the brain.

Allows blood from either vessels to be distributed to any part of both cerebral hemispheres.

Contributing arteries include:

Anterior communicating artery.

Anterior cerebral arteries. (Right and left)

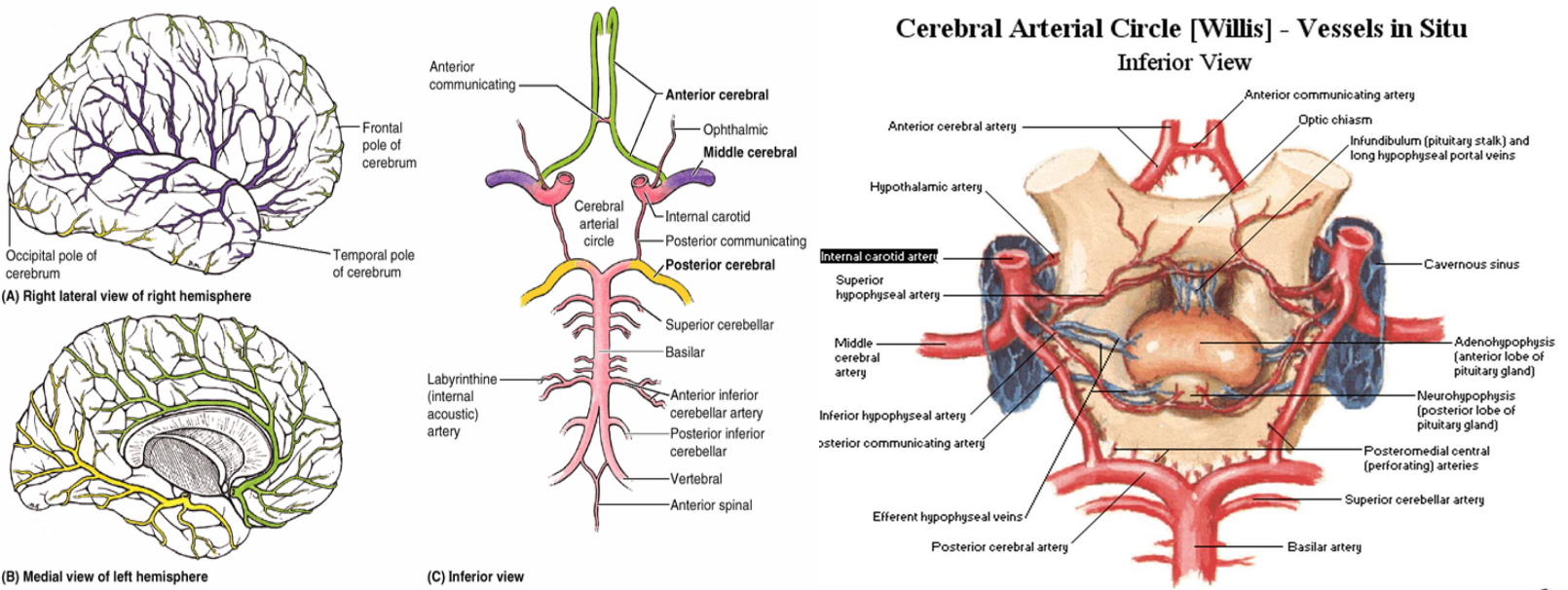
Internal carotid arteries (Right and left)

Posterior communicating arteries (Right and left)

Posterior cerebral arteries (Right and left)

Cerebral Arterial Circle [Willis] - Vessels in Situ

Inferior View



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